

CareConnect: Patient-Provider Electronic Health Record

Enterprise-Grade Full-Stack EHR System with Real-Time Drug-Allergy Safety Engine, Conflict-Free Scheduling, Medical Report Vault, and HIPAA Audit Governance

LIVE WEB APPLICATION

<https://careconnect-ehr.vercel.app/>

React 19 + Vite + Tailwind CDN

LIVE REST API & SWAGGER

<https://careconnect-backend-uim9.onrender.com>

Spring Boot 3 + Java 17 + Docker

SOURCE CODE REPOSITORY

<https://github.com/faizan0264/careconnect-ehr>

Oracle 19c/21c DDL + Monorepo

1. Problem Statement & Clinical Motivation

Preventable medical errors and scheduling friction cost hundreds of thousands of lives and billions of dollars annually.

Medical Errors & Siloed Charts

- Over 250,000 deaths annually occur from preventable adverse drug events (ADEs).
- Physicians lack instantaneous allergy cross-checks at the moment of electronic prescribing.
- Crucial contraindications (e.g., Penicillin allergies) are buried in fragmented paper/PDF files.

Patient Disempowerment

- Patients lack real-time digital access to their longitudinal health history and visit plans.
- Unable to securely transmit outside radiology scans or prior hospital records to treating doctors.
- Significant friction in requesting prescription refills leads to therapy discontinuation.

Scheduling Chaos & Collisions

- Clinic double-booking causes severe physician burnout and multi-hour patient waiting times.
- Traditional booking forms fail to check doctor availability before committing slots.
- Lack of immutable HIPAA audit trails creates severe regulatory non-compliance risks.

2. The CareConnect Solution: 3 Unified Portals

Connecting patients, physicians, and administrators within a synchronized, highly secure ecosystem.

Doctor Clinical Workstation

- Active clinic consultation queue with 1-click chart access.
- Full SOAP clinical encounter notes (Subjective, Objective, Assessment, Plan).
- Comprehensive vitals history tracking.
- Computerized Physician Order Entry (CPOE) for Labs and Imaging.
- e-Prescribing with real-time Allergy Safety Engine.
- Diagnostic scan inspection and report review.

Patient Self-Service Portal

- Longitudinal health summary and vital signs history.
- Digital After-Visit Summary (AVS) with 1-click physical printing.
- Conflict-free appointment booking with real-time schedule busy-slot detection.
- 1-click prescription refill requests with pharmacy tracking.
- Medical Report Vault for outside record upload.
- Self-registration with automatic MRN allocation.

Admin Governance & HIPAA

- Hospital-wide census and clinical volume analytics.
- User provisioning: add verified physicians and nurses.
- Staff account deactivation and credential updates.
- Immutable HIPAA Audit Trail capturing every access, order, and document event with IP and timestamp.
- Strict zero-phantom-login enforcement.

3. System Architecture & Engineering Stack

Built on modern enterprise standards ensuring separation of concerns, high throughput, and zero downtime.

Client Tier: React 19 + Vite 6

- **React 19** component-driven UI with modern hooks.
- **Tailwind CSS** clinical design system.
- **EhrContext** centralized state management.
- **HTML5 FileReader** Base64 document encoding.
- **LocalStorage** dual-sync for offline resilience.
- **Vercel Edge Cloud** global CDN hosting.

Server Tier: Spring Boot 3.3.4

- **Java 17+ (LTS)** enterprise object-oriented runtime.
- **Spring MVC** REST API controllers.
- **Spring Data JPA** repository abstractions.
- **Hibernate 6** ORM & schema management.
- **OpenAPI 3.0 / Swagger UI** interactive API sandbox.
- **Render Docker Container** microservice.

Data Tier: Oracle 19c/21c DDL

- **8 Relational Tables** with full foreign keys.
- **ON DELETE CASCADE** referential rules.
- **CHECK & UNIQUE Constraints** on roles & MRNs.
- **B-Tree Indexes** on dates, MRNs, timestamps.
- **CLOB Storage** for encrypted Base64 documents.
- **H2 in-memory** zero-dependency fallback.

4. Star Features: Clinical Safety & Conflict-Free Scheduling

Proprietary clinical intelligence algorithms protecting patients and optimizing clinic operations.

Drug-Allergy Safety Engine

- **Real-Time Interception:** When a doctor prescribes a drug, the engine cross-references patient allergies.
- **Contraindication Modal:** Prescribing Amoxicillin to a Penicillin-allergic patient triggers an immediate high-priority warning.
- **Override Accountability:** Requires formal clinical justification ('Benefit outweighs risk') and signed acknowledgment.
- **Audit Trail:** Logged to HIPAA audit table.

Schedule Collision Prevention

- **Busy-Slot Detection:** Real-time check queries doctor appointments on selected date.
- **Visual Busy Alerts:** Occupied slots (e.g. 10:30 AM) are highlighted in rose with 'Busy ●' tag.
- **Booking Lock:** Submit button disables automatically with conflict message.
- **API Isolation:** Backend enforces concurrency lock returning HTTP 409 if double-booked.

Medical Report Vault

- **Dual-Access:** Both patients and doctors upload files up to 15MB.
- **HTML5 Ingestion:** Drag-and-drop file upload with Base64 CLOB persistence.
- **In-Browser Preview:** Inspect radiology scans without third-party tools.
- **Clear Attribution:** Tagged as 'Physician File' vs 'Patient Record'.
- **Downloads:** One-click diagnostic downloads.

5. Live Deployment, Governance & Evaluator Credentials

Live on Vercel and Render with interactive Swagger API docs and complete defense credentials.

Doctor Persona Credentials

- **Portal:** Doctor Portal
- **Username:** **dr_smith** (or *dr.sharma*)
- **Password:** **password123** (or *Doctor#2026*)
- **Specialty:** Internal Medicine & Pulmonology
- **Key Demo:** Open chart for John Doe, review vitals/SOAP, prescribe Amoxicillin to demonstrate Allergy Interception!

Patient Persona Credentials

- **Portal:** Patient Portal
- **Username:** **patient1** (or *john_doe*)
- **Password:** **Patient#2026** (or *password123*)
- **MRN:** MRN-2026-0042 (Allergy: Penicillin)
- **Key Demo:** Try booking Dr. Sarah Smith at 10:30 AM on Oct 14 to show Conflict Alert; upload outside lab report!

Administrator Credentials

- **Portal:** Admin Console
- **Username:** **admin**
- **Password:** **Admin#2026**
- **Key Demo:** Open HIPAA Audit Trail tab to show real-time immutable logging of all chart views, orders, and uploads with IP address and timestamps.